**Personal details and date of CV**

- Surname: Mao
- First name: Zhu
- ORCID: 0000-0002-8464-2558
- Date of CV: 10 Nov. 2026

**Degrees**

- PhD in Photogrammetry and Remote Sensing, State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, Jun. 2023
- MSc in Surveying and Mapping Engineering, LIESMARS, Wuhan University, Wuhan, Jun. 2018
- BSc in Geographic Information Science, Nanjing University of Information Science & Technology (NUIST), Nanjing, Jun. 2016

**Current employment**

- Postdoctoral researcher, Department of Forest Sciences, University of Helsinki (research group ForTech), Jan. 2024–present
  - Stage of academic research career: Stage II
  - Research on AI for forest, remote sensing, and precision forestry: developing methods that characterize boreal forest structure and soils from multi-source remote sensing data, including UAV, aerial and laser scanning data, analyzing logging trail networks from forest-machine data, and translating the results into support for forest operations.

**Previous work experience**

- Editorial Assistant (part-time), *Geo-spatial Information Science* (GSIS), Wuhan, Jan. 2021–Dec. 2023
  - Open-access international journal
- Doctoral researcher, LIESMARS, Wuhan University, Wuhan, Sep. 2019–Jun. 2023
  - Research on oblique photogrammetric 3D reconstruction and AI-based scene understanding.
- Business Analyst, IBM CIO Identity team, Chengdu, Jul. 2018–Aug. 2019
  - Worked in an international team on enterprise identity and access management.

**Research funding and grants**

- GeoAI explained soil mapping of Finnish forests (GeoAISoil), Finnish Reseach Impact Foundation, May 2025–Apr. 2027
  - Role: key researcher (PI: Prof. J. Uusitalo); in collaboration with Arbonaut
- Precision Forest Harvesting, project funded by industry, Jan. 2024–Jun. 2025
  - Role: key researcher (PI: Prof. J. Uusitalo)

**Research output**

- Publications: 14 peer-reviewed journal articles on AI-based remote sensing and photogrammetry, with applications in forestry and geoinformation science.
- Open-source software
  - MT-GAT: multi-task learning graph attention network for predicting soil organic layer thickness and organic matter in boreal forests, 2026
  - LTA-Logging Trail Analyzer: a Python-based workflow for deriving and evaluating logging trail networks from forest-machine GNSS data, 2026

  - SR-Supporting-ITD: individual tree detection and canopy stratification from half-meter aerial data, 2025
- Open datasets
  - URFD: slender urban road facility extraction from oblique aerial images, 2023
  - GFSD: glass facade segmentation from oblique aerial images, 2022

**Research supervision and leadership experience**

- **Co-supervisor** of one doctoral researcher and two MSc students, University of Helsinki

**Teaching merits**

- **Guest lecturer**, MSc courses in Forest Sciences, University of Helsinki, 2024–2026
  - FOR-265 *Terramechanics and Trafficability Prediction*, 2026
  - FOR-263 *Operations Research in Forest Resource Management*, lecture "Computer Vision in Forestry", 2024, 2026
  - FOR-280 *Precision Harvesting*, 2025
  - FOR-018 *Introduction to the MSc Programme in Forest Sciences*, 2025, 2026

**Other key academic merits**

- Lead Guest Editor, Special Issue "Remote Sensing of Individual Trees: Delineation, Characterization, and Applications", *Remote Sensing* (MDPI), 2026
- Peer reviewer for international journals:
  - including *ISPRS Journal of Photogrammetry and Remote Sensing* (ISPRS P&RS), *International Journal of Applied Earth Observation and Geoinformation* (JAG), *Scientific Data* (SD), *Remote Sensing* (RS) and *Geo-spatial Information Science* (GSIS).
- Oral presentations
  - "Mapping tree-level structural diversity for precision harvesting operations: A comparison of open-source and drone datasets", Forest Science Day (Metsätieteiden päivä) 2025, Helsinki, Nov. 2025
  - "Deep-learning-based individual tree detection for tomorrow's precision harvesting operations", FORMEC 2025, Joensuu, Jun. 2025, slides online.

**Scientific and societal impact**

- **Open science:** I release the code and data behind my research. The methods SR-Supporting-ITD, MT-GAT and LTA and the benchmark datasets URFD and GFSD are openly available on GitHub (see Research output), so that other researchers can reproduce and build on them.
- **Transfer to practice:** In the GeoAISoil project, I work with Arbonaut, a forest information company, to bring AI-based soil property mapping into forest planning and harvesting operations. The LTA tool supports the evaluation of logging trail networks from forest-machine data.
- **Communication with forest professionals:** I present my research to the forest sector at national and international meetings (Metsätieteiden päivä 2025, FORMEC 2025) and in MSc courses for future forest professionals.

**Other**

- **Technical skills:** Python (PyTorch, scikit-learn, GDAL); ArcGIS, QGIS, CloudCompare, Bentley ContextCapture, LaTeX; Linux, CSC supercomputing, Git
- **Languages:** Chinese (native), English (fluent)